

Analytical Modeling of a Soft Pneu-Net Actuator Subjected to Planar Tip Contact

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Abstract—Soft actuators are compliant structures that are generally made of elastomers and generates large deformation. The behavior of these structures cannot be estimated accurately using infinitesimal strain theories. The objective of soft robotics applications is the controlled large deformation of these structures. In this article, we propose an analytical model for a pneu-net soft actuator. The model is based on the Euler–Bernoulli finite strain hyperelastic thin cantilever beam theory. The deformation of the air chambers is modeled using finite strain membrane theory. The analytical model is developed for two different states of the actuator: 1) free space; and 2) when the actuator was subjected to tip contact. The proposed theoretical model predicts the deformation and force characteristics of the actuator for the grasping state. The theoretical formulation of the developed model is different from previously developed infinitesimal strain models for the actuator, as it considers the axial stretch and forces applied to the actuator. In addition, it can be theoretically implemented on similar structured actuators for various applications. The theoretically calculated deformation and force characteristics of different actuators are compared with the finite element (FE) model and experimental characteristics. The results suggest that the proposed model can predict the actuator deformation and force characteristics as accurately as the FE model, but the computation time of the proposed model is less than 1% that of the FE model. The proposed model is further implemented on a three-finger gripper to predict the air pressure required for a stable grasp of different objects and is validated experimentally.

Index Terms—Analytical modeling, block force, contact, grasping, soft pneu-net actuator.

I. INTRODUCTION

SOFT actuators have the potential to address the challenges of soft body interactions during the handling of soft objects [1]. Such interactions are generally encountered while handling

products in food and agriculture industry [2]–[4], handling living creatures during oceanographic exploration [5], [6], and in health care applications [7]. To design and manufacture soft actuators, accurately and efficiently, their physics needs to be explored [8]. For the efficient design of actuators, accurate models are required to examine their deformation behavior [1]. Recently, pneu-net soft actuators have gained significant attention owing to their fast actuation in bending [9], [10]. These actuators have numerous applications [11]–[13]. To improve the design and performance of these actuators, various approaches have been presented [14]–[21].

In the past, researchers dealt with the modeling aspect of soft actuators [22]–[28]. The existing models of these actuators are Euler–Bernoulli beam-based constant curvature [29]–[33], piecewise constant curvature [34], [35], Cosserat rod approach [36], lumped parameter [37], and finite element (FE) models [38]–[40]. FE-based simulations are widely used for the structural optimization of these actuators [41]–[43]. Analytical models for soft actuators have also been presented by researchers. Polygerinos *et al.* [40] presented a quasi-static analytical and FE model for a fiber-reinforced actuator. Sadati *et al.* [35] presented a geometric deformation constant and variable curvature model for a braided continuum manipulator. Paoletti *et al.* [26] presented an analytical model of a soft actuator to grasp soft and stiff objects. A recent study discussed a general approach for the analytical modeling and design of soft extending and contracting devices [28]. Connolly *et al.* [27] presented an analytical model of an actuator for trajectory tracking.

Analytical models of pneu-net actuators were presented in [44]–[49]. The authors in [30] and [33] presented Euler–Bernoulli beam-based models for a pneu-net actuator. Cao *et al.* [44] developed a constant-curvature model of a pneu-net actuator for model-based control of the actuator. Based on the constant curvature assumption, Liu *et al.* [45] calculated the bending angle and the tip forces of a pneu-net actuator by considering the deformation of air chambers. Majidi *et al.* [46] presented an analytical model based on the principles of elasticity and friction for a locomotion soft robot. Wang *et al.* [47] presented an analytical model of a pneu-net actuator based on the deformation of air chambers and serial robot assumption. Fang *et al.* [48] presented a geometry-based framework for computing the deformation of a pneu-net actuator within the range of linear elasticity. Xu *et al.* [49] developed a mathematical model of a pneu-net actuator based on the multiple point contact approach for contact between air chambers. Payrebrune *et al.* [50] presented an Euler theory of elastica-based rod model

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for a pneu-net soft bending actuator. Gu *et al.* [51] presented an analytical model based on rod theory for a pneu-net actuator with oblique chambers. Wang *et al.* [52] presented a simplified line-segment model for a pneu-net actuator using an optimization-based parameter identification method. Zhou *et al.* [53] presented an analytical model for a soft pneumatic actuator used for bionic robotic fish with symmetrical chambers.

The analytical modeling of these actuators is in its developmental phase and is still not sufficiently mature. Despite the aforementioned developments, well-understood analytical models of the actuators are yet to be developed because their intrinsic deformation is continuous, complex, and highly compliant. Most analytical models considered the free space deformation but did not account for external contact. A brief analysis of the features of existing analytical models of soft pneu-net actuators is presented in Table IV. From the analysis, as an outcome, the modeling of air chambers is not yet well-investigated. The actuators contain many passive degrees of freedom, and when driven at a low input pressure, the available pneumatic power cannot compensate for gravity loading [23]. FE-based simulations have been widely used to model and simulate the behavior of a pneu-net actuator and they work well for reproducing the deformation and stress of the actuator. However, for real time applications or application requiring fast calculations, computationally efficient models are required. Therefore, it is necessary to develop methods to account for these factors. This study is focused on the development of an analytical and computationally less expensive model for a pneu-net actuator, which is based on the Euler–Bernoulli finite strain beam model. The incompressible neo-Hookean material model is considered to account for the hyperelastic material behavior of the actuator. The deformation of air chambers when subjected to pressure was modeled using the finite strain membrane theory. The analytical model was developed for two different states of the actuator: 1) free space; and 2) when the actuator is subjected to tip contact. The developed model has the following key features:

- 1) The theoretical formulation is different from previously developed infinitesimal strain models for the actuator because it considers axial stretch, and tip contact. It can be easily used in various applications by modifying the boundary conditions of the governing equation.
- 2) The finite strain model for the air chambers is based on membrane theory and contact mechanics approach.
- 3) It can be applied to a class of pneumatic soft bending actuators with rectangular base and different air chamber geometries by modifying the air chamber modeling.
- 4) When compared with the previously developed constant curvature and piecewise constant curvature models, the developed model simultaneously predicts the deformation and force characteristics of the actuator.
- 5) The deformation and force characteristics predicted by the theoretical model are similar to those of the FE model, but the computation time required by the theoretical model was less than 1% of that of the FE model.

The modeling approach for the actuator subjected to tip contact was further applied to a three-finger gripper to predict the air pressure required for a stable grasp of different objects.

Analytical computations were performed for free space, blocked force, and grasping state of the actuator and are experimentally validated.

The presented work is an extension of our earlier work [54]. Significant progress has been made to the earlier work and the presented work has several key additions. The model is extended to study the grasping behavior of the actuator and to predict the deformed characteristics of nonuniform geometry actuator. Based on the contact mechanics approach and finite strain membrane theory, a new model is proposed for the air chambers. The actuator model is further tested on actuators made of different materials and geometric structures. Additional experiments were conducted for grasping state, actuators subjected to different offset heights at the tip, and to determine the material and frictional properties of the actuators.

The rest of this article is organized as follows. The theoretical background of the modeling is presented in Section II, followed by the derivation of the governing equations and numerical solution procedure in Section III. The basis of the analytical computations, FE simulations, and the procedure for conducting experiments for different states of the actuator are presented in Section IV. Section V presents discussions on the analytical, FE simulation, and experimental results. Finally, Section VI concludes this article.

II. THEORETICAL FORMULATION

A. Actuator Design and Theoretical Assumptions

The soft actuator design is based on the pneu-net morphology of the elastomer robot. A detailed discussion on actuator design was presented in [55]. The casting process was employed to fabricate the actuator using silicone rubber. Certain assumptions were made while modeling the actuator. The pressure is assumed to be uniform throughout the actuator, and the friction at the contact between adjacent chambers is not considered in this study. As shown in Fig. 1, the cover pad at the base of the actuator is modeled as a hyperelastic cantilever beam with a rectangular cross section. The air chambers are modeled as independent elements that apply a distributed and an eccentric axial force on the cover. The distributed force is due to the weight of air chambers. The eccentric axial force is induced by the air pressure, and it is predicted using the deformation of air chambers side walls. The bending stiffness of air chambers is not considered. The model assumes that the applied air pressure is adequate and all air chambers are in contact. The modeling approach considers only flat geometry chambers as the initial undeformed state.

B. Modeling of Air Chambers

The contact force developed between the air chambers is derived from the deformation of air chambers. The air chambers are modeled as a flat membrane using the large strain membrane theory [56], [57]. The side wall (in the YZ plane) of the air chambers is modeled as an initially flat membrane with clamped edge condition and subjected to lateral pressure. The deformation of the remaining walls is not considered in this study. When

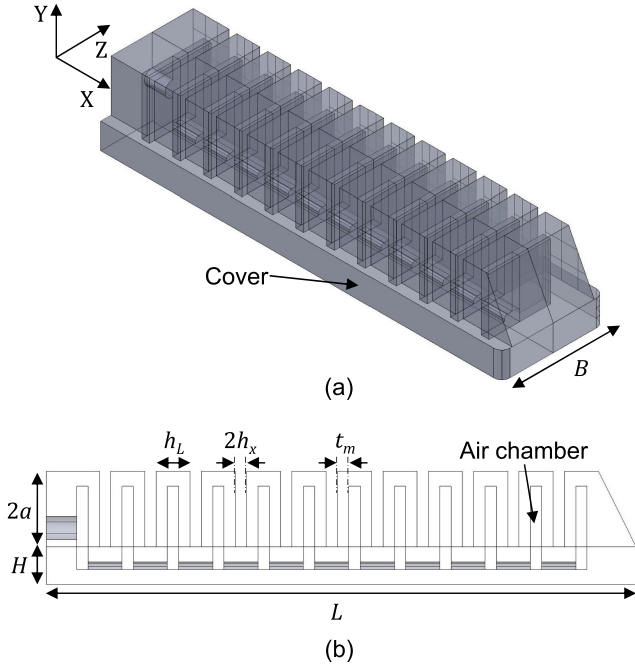


Fig. 1. Design of pneu-net actuator. (a) Isometric view. (b) Section view along the length.

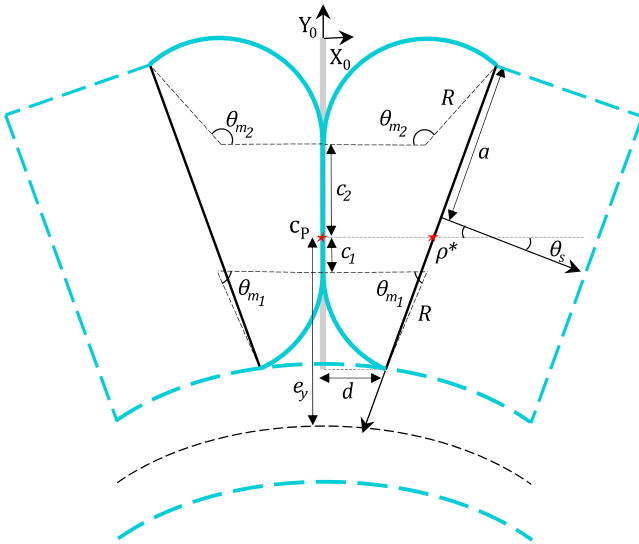


Fig. 2. Schematic of deformed air chambers in contact. The solid aquamarine color lines represents the deformed membrane profile.

subjected to internal pressure, the membranes get inflated and come into contact with the adjacent membranes, as shown in Fig. 2. The deformation of a flat square membrane subjected to an inclined contact plane is expressed by the equation (for nomenclature refer Table I and detailed derivation refer to the Supplementary Material, Section-1)

$$1 - \lambda_m^{-6} = \frac{p(d + 2a \sin \theta_s)}{4\mu t_m \sin^2 \left(\frac{\theta_{m2}}{2} \right)} \quad (1)$$

where

$$\lambda_m = \cos \theta_s + \frac{d + 2a \sin \theta_s}{2(a + \rho^*)} \left[\frac{\theta_{m2}}{\sin^2 \left(\frac{\theta_{m2}}{2} \right)} - 2 \cot \left(\frac{\theta_{m2}}{2} \right) \right]. \quad (2)$$

The contact region between the air chambers is assumed to be elliptical and the minor and major axes are defined as c_y and c_z . From the deformed geometry, c_y and c_z can be expressed as

$$c_y = c_1 + c_2 \quad (3)$$

$$c_z = \frac{c_y b (a - \rho^*)}{a^2}. \quad (4)$$

The contact widths c_2 and c_1 can be expressed as

$$c_2 = (a + \rho^*) \cos \theta_s - (d + 2a \sin \theta_s) \cot \left(\frac{\theta_{m2}}{2} \right) \quad (5)$$

$$c_1 = 2a\lambda_m - R \left[\cos^{-1} \left(1 - \frac{d}{R} \right) + \theta_{m2} \right] - c_2. \quad (6)$$

The radius of curvature R can be expressed as

$$R = \frac{d + 2a \sin \theta_s}{2 \sin^2 \left(\frac{\theta_{m2}}{2} \right)}. \quad (7)$$

The contact area A_c for the elliptical region can be written as

$$A_c = \frac{\pi}{4} c_y c_z. \quad (8)$$

The contact force F_p generated by the air chambers can then be calculated as

$$F_p = p A_c. \quad (9)$$

The force F_p induced a bending moment M_m on the segment $2h_x$ (initially, $d = h_x$) and it can be written as

$$M_m = F_p e_y \quad (10)$$

where e_y is the eccentricity and from the deformed geometry (see Fig. 2), it can be defined as

$$e_y = \left(\frac{H}{2} + a - \rho^* \right) \cos \theta_s. \quad (11)$$

The bending moment M_m induced on segment $2h_x$ is one of the driving forces that causes the bending deformation of the cover pad. The gross effect of this driving force is also dependent on other design parameters, i.e., density and length of the air chambers h_L . For segment X_{L_i} the bending deformation caused by M_m is equivalent to the bending deformation caused by an unknown moment M_{p_i} which is applied on the span of X_{L_i} , as represented in the schematics Fig. 3(a) and (b). The unknown moment M_{p_i} can be calculated by equating the change in strain energy caused by M_m and M_{p_i} , and in the final form it has the following expression:

$$M_{p_i} = \left(\frac{2h_x M_m^2}{2h_x + h_L} \right)^{\frac{1}{2}}. \quad (12)$$

TABLE I
LIST OF SYMBOLS

Symbol	Description	Symbol	Description
λ_m	Stretch in the membrane	M_{F_i}	Bending moment induced by point forces
d	Offset between membrane and substrate, initially $d = h_x$	θ	Rotation angle of deformed plane of the beam
p	Air pressure	u_0, v_0	Displacement components of beam's material line
$2a$	Height of air chamber or width of membrane	λ	Stretch of beam's material line
$2b$	Breadth of air chamber or length of membrane	κ	Curvature of deformed midplane of the beam
$2\theta_s$	Inclination between adjacent chambers	ds	Deformed length of the beam
μ	Material shear modulus	U	Strain energy density function
t_m	Membrane thickness	I_1	First strain invariant
θ_{m_2}	Deformed membrane exterior angle	g_x, g_y	Distributed forces per unit reference length
θ_{m_1}	Deformed membrane interior angle	T	Axial stress resultant
ρ^*	Undeformed location of contact initiation point C_P	M	Bending moment acting on deformed surface
c_1, c_2	Interior and exterior contact widths	m	Distributed bending moment
c_y, c_z	Minor and major axis of elliptical contact region	P	Point force along X direction or component of stress resultant
R	Radius of curvature	Q	Point force along Y direction or component of stress resultant
A_c	Contact area between air chambers	L	Length of actuator
F_p	Contact force between air chambers	C_1, C_2	Constants of integration
$2h_x$	Initial gap between the air chambers	B	Width of actuator cover
M_m	Bending moment induced on segment $2h_x$ by force F_p	H_{ext}, F_{ext}	Horizontal and vertical tip forces
X_{L_i}	Length of beam segments	G	Integral functional constraint
M_{p_i}	Bending moment induced on each beam segment (X_{L_i})	Λ	Lagrange multiplier
e_y	Eccentricity of axial force F_p	y_c	Object offset
H	Thickness of actuator cover	m_h	Mass of grasping object
h_L	Length of air chamber (along X axis)	μ_f	Coefficient of friction (COF)
M_T	Equivalent bending moment	E	Modulus of elasticity
x_{t_i}	Distance of beam segment from free end	ν	Poisson's ratio
M_{g_i}	Bending moment induced by distributed forces	g	Gravitational acceleration

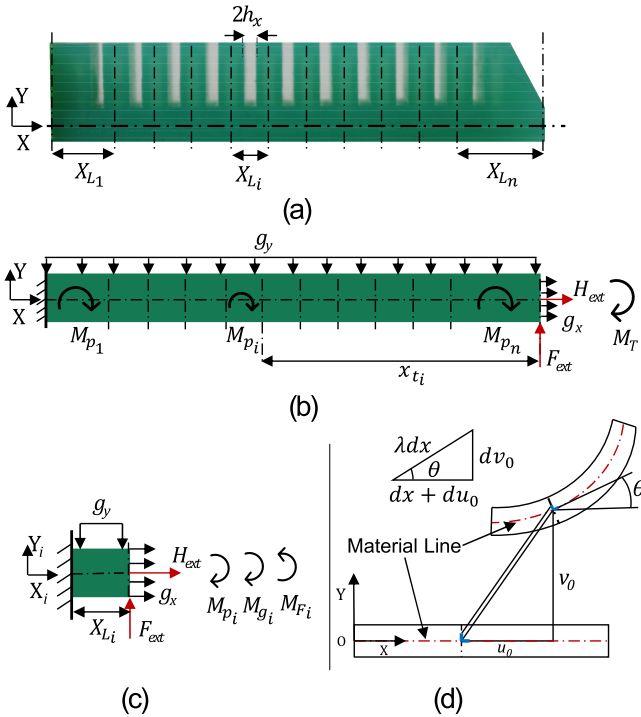


Fig. 3. Schematic of (a) beam segments, (b) free body diagram of actuator cover in undeformed state, (c) free body diagram of a beam segment in undeformed state, and (d) geometry of deformed and undeformed configuration of plain strain hyperelastic beam.

For a uniform geometry actuator, the bending moment M_{p_i} is uniform throughout the actuator, and in the later derivations, it is represented as $M_{p_i} = M_T$. The actuator has a pneu-net geometry, and the application point of force F_p is at an offset of e_y from the actuator cover. Thus, the bending moment M_{p_i}

is the primary component of force F_p . The effect of the axial component of force F_p is not significant when compared with the effect of bending moment; hence, it is not considered in further discussion.

C. Bending Moment for Nonuniform Air Chambers

For a nonuniform geometry actuator, the deformation and force characteristics of each beam segment are evaluated separately. The free body diagram of a beam segment in the undeformed state is represented in Fig. 3(c). At the free end, the bending moments M_{g_i} and M_{F_i} are introduced by the distributed and tip forces and can be written as

$$M_{g_i} = g_y \frac{x_{t_i}^2}{2} \quad (13)$$

$$M_{F_i} = F_{ext} \cdot x_{t_i} \quad (14)$$

For free space deformation state, the equivalent bending moment M_T at the free end is considered as

$$M_{T_i} = \max(M_{p_i}, M_{g_i}) \quad (15)$$

In the presence of tip forces, the equivalent bending moment M_T is considered as

$$M_{T_i} = M_{p_i} + M_{g_i} - M_{F_i} \quad (16)$$

The expressions for M_T are different for the free space and tip contact states because the application of the bending moment M_p is dependent on other forces. The beam segments are serially connected to each other. The final deformation of the actuator is derived by transforming the individual beam segment deformation to a common inertial frame. The modeling of the actuator cover as a thin rectangular cross-sectional beam is discussed in the following section:

D. Modeling of Actuator Cover

The modeling of actuator cover followed the plane strain hyperelastic thin beam model developed by He *et al.* [58]. The geometries of deformed and undeformed configurations of the cover are shown in Fig. 3(d). The rotation angle θ and stretch λ are defined as

$$\theta = \arctan \frac{v'_0}{1 + u'_0} \quad (17)$$

$$\lambda = \sqrt{(1 + u'_0)^2 + v'^2_0} \quad (18)$$

where $(\cdot)'$ represents the derivative with respect to coordinate X . The curvature of the deformed midplane κ is defined as

$$\kappa = \frac{d\theta}{ds} = \frac{d\theta}{\lambda dX} = \frac{\theta'}{\lambda} \quad (19)$$

where ds is the deformed length of line element dX on the geometrical midplane. The strain energy density function U for incompressible neo-Hookean material [59] is defined as

$$U(I_1) = \frac{1}{2}\mu(I_1 - 3). \quad (20)$$

He *et al.* [58] developed the following Euler-Lagrangian equations for hyperelastic beams:

$$\left(T \cos \theta + \frac{M' + m}{\lambda} \sin \theta \right)' + g_x = 0 \quad (21)$$

$$\left(T \sin \theta - \frac{M' + m}{\lambda} \cos \theta \right)' + g_y = 0. \quad (22)$$

The axial stress resultant T and bending moment M are defined as

$$\frac{T}{\mu BH} = \lambda - \frac{1}{\lambda^3} - \frac{\kappa^2 H^2}{\lambda^7} \quad (23)$$

$$\frac{M}{\mu BH^3} = \frac{\kappa}{3\lambda^6}. \quad (24)$$

The boundary conditions associated with these equations can then be derived as

$$\begin{aligned} P &= P_X \text{ or } \delta u_0 = 0, \text{ at } X = 0 \text{ and } X = L \\ Q &= Q_X \text{ or } \delta v_0 = 0, \text{ at } X = 0 \text{ and } X = L \\ M &= M_X \text{ or } \delta \theta_0 = 0, \text{ at } X = 0 \text{ and } X = L. \end{aligned} \quad (25)$$

The components P and Q of the stress resultant vector are defined as

$$P = T \cos \theta + \frac{M' + m}{\lambda} \sin \theta \quad (26)$$

$$Q = T \sin \theta - \frac{M' + m}{\lambda} \cos \theta. \quad (27)$$

We reformulated these equations to capture the deformation of the actuator and developed the governing equations for the free space and tip contact states of the actuator.

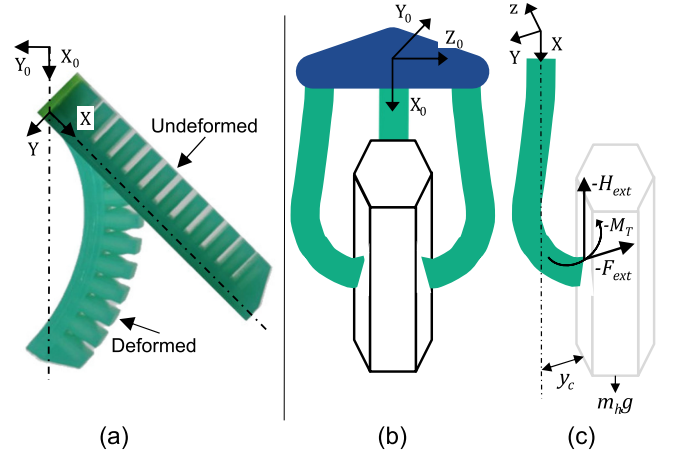


Fig. 4. Different states. (a) Actuator in free space, (b) schematic of deformed state of a three-finger gripper grasping a hexagonal cylinder, and (c) free body diagram of single finger during grasping.

III. FORMULATION OF GOVERNING EQUATIONS AND NUMERICAL SOLUTION

A. Free Space Deformation

In this state, the actuator is subjected to gravitational force and internal pressure. As a general case, an initially inclined state of the actuator is considered for the study, as shown in Fig. 4(a). The boundary conditions for this state are defined as

$$u_0 = 0, v_0 = 0, \text{ and } \theta_0 = 0, \text{ at } X = 0$$

$$P = 0, Q = 0, \text{ and } M = M_T, \text{ at } X = L. \quad (28)$$

Integrating (21) and (22) with respect to X yields the following:

$$T \cos \theta + \frac{M' + m}{\lambda} \sin \theta = -g_x X + C_1 \quad (29)$$

$$T \sin \theta - \frac{M' + m}{\lambda} \cos \theta = -g_y X + C_2. \quad (30)$$

The constants of integration C_1 and C_2 are evaluated by substituting the boundary conditions in (29) and (30). Here, g_x and g_y are components of the gravitational force per unit length along X - and Y -directions, respectively, and $m = 0$, as the distributed moment is not applied. After substituting the values of C_1 , C_2 , T (23) and M (24) into (29) and (30), the equations are reduced to the following governing equations:

$$\frac{1}{\lambda} \left(\frac{H^2 \kappa}{3\lambda^6} \right)' = \frac{g_x(L - X)}{\mu BH} \sin \theta - \frac{g_y(L - X)}{\mu BH} \cos \theta \quad (31)$$

$$\lambda - \frac{1}{\lambda^3} - \frac{\kappa^2 H^2}{\lambda^7} = \frac{g_x(L - X)}{\mu BH} \cos \theta + \frac{g_y(L - X)}{\mu BH} \sin \theta. \quad (32)$$

The third boundary condition $M = M_T$ at $X = L$ (28) is modified using (24) as

$$\theta' = \frac{3M_T \lambda^7}{\mu BH^3}. \quad (33)$$

To consider the terminal forces in the free space state, the boundary conditions at $X = L$ are modified as: $P = H_{\text{ext}}$ and $Q = F_{\text{ext}}$.

B. Actuator Subjected to Tip Contact

In this state, the actuator is subjected to gravitational force, internal pressure, and constraints at the tip. As a general case, the state of the actuator while grasping an object in pinching mode is considered for study. A stable grasp between a three-finger gripper and a hexagonal cylinder is considered, as shown in Fig. 4(b). The local coordinates for each finger are attached at the fixed end of the finger, and the global coordinates for the gripper are attached at the center of gripper base. To grasp an object, the actuator should apply a defined amount of force and maintain a defined geometric configuration with respect to the object. The amount of force applied by the actuator depends on the weight of the object and contact conditions (static case). The geometric configuration depends on the geometry and position of the object with respect to the actuator. As shown in Fig. 4(b) and (c), for the grasping state, the actuator applied a lifting force H_{ext} along the X -direction and maintained an offset y_c at the tip along the Y -direction. The boundary conditions for this state are defined as

$$u_0 = 0, v_0 = 0, \text{ and } \theta_0 = 0, \text{ at } X = 0$$

$$P = H_{\text{ext}}, v_0 = y_c, \text{ and } M = M_T, \text{ at } X = L. \quad (34)$$

The boundary condition at $X = L$ along the Y -direction is defined by a functional-type constraint as

$$\int_0^L G(u'_0, v'_0) dX - y_c = \int_0^L \lambda \sin \theta dX - y_c = 0. \quad (35)$$

The effect of the integral functional constraint $G(u'_0, v'_0)$ is incorporated into the Euler–Lagrangian equations (21) and (22) using the Lagrange multiplier Λ . The modified governing equations can then be written as

$$\left(T \cos \theta + \frac{M' + m}{\lambda} \sin \theta \right)' - \left(\Lambda \frac{\partial G}{\partial u'_0} \right)' + g_x = 0 \quad (36)$$

$$\left(T \sin \theta - \frac{M' + m}{\lambda} \cos \theta \right)' - \left(\Lambda \frac{\partial G}{\partial v'_0} \right)' + g_y = 0. \quad (37)$$

The partial differentials of functional $G(u'_0, v'_0)$ have the values

$$\frac{\partial G}{\partial u'_0} = 0, \text{ and } \frac{\partial G}{\partial v'_0} = 1. \quad (38)$$

From variational calculus, the natural boundary condition at $X = L$, along the Y -direction is defined as

$$T \sin \theta - \frac{M' + m}{\lambda} \cos \theta - \Lambda = 0. \quad (39)$$

From (27) and (39), it is concluded that for this state, the Lagrange multiplier Λ represents the component Q of stress resultant vector along the Y -direction. The stress resultant component Q for this state is the reaction force generated by the actuator at the tip ($X = L$) along the Y -direction. If the coefficient of friction (COF) μ_f at the contact region is known, then from the boundary condition $P = H_{\text{ext}}$ at $X = L$, the value of the

Lagrange multiplier should be $\Lambda \geq H_{\text{ext}}/\mu_f$, using Coulomb's law of friction. Integrating (36) and (37) with respect to X yields the following:

$$T \cos \theta + \frac{M' + m}{\lambda} \sin \theta = -g_x X + C_1 \quad (40)$$

$$T \sin \theta - \frac{M' + m}{\lambda} \cos \theta - \Lambda = -g_y X + C_2. \quad (41)$$

The constants of integration C_1 and C_2 are evaluated by substituting the boundary conditions (34) into (40) and (41) as

$$C_1 = H_{\text{ext}} + g_x L$$

$$C_2 = g_y L. \quad (42)$$

Substituting the values of C_1 , C_2 , T (23), and M (24) into (40) and (41) yields the following governing equations:

$$M' = \lambda (H_{\text{ext}} + g_x (L - X)) \sin \theta - \lambda (\Lambda + g_y (L - X)) \cos \theta \quad (43)$$

$$\lambda - \frac{1}{\lambda^3} - \frac{\kappa^2 H^2}{\lambda^7} = \frac{H_{\text{ext}} + g_x (L - X)}{\mu B H} \cos \theta$$

$$+ \frac{\Lambda + g_y (L - X)}{\mu B H} \sin \theta. \quad (44)$$

Equation (33) is also a part of the set of governing differential equations, when expressed in first-order form. The local material line coordinates in the deformed state are determined using the following:

$$X = \int_0^L \lambda \cos \theta dX \quad (45)$$

$$Y = \int_0^L \lambda \sin \theta dX. \quad (46)$$

Equation (46) also adds an equation to the set of first-order differential equations as:

$$Y' = \frac{dY}{dX} = \lambda \sin \theta. \quad (47)$$

For the grasping state shown in Fig. 4(b), it is assumed that the deformations of all fingers are identical. The global material line coordinates, shown in Fig. 3(d), of the fingers are determined by transforming the coordinates from local to global frame.

Finally, by solving the governing differential (33), (43), and (47) and algebraic (44) equations, we can calculate the axial stretch and slope at material line coordinates in the deformed state. The material line coordinates in the deformed state can be calculated from (45) and (46) by substituting the values of the axial stretch and slope.

C. Numerical Solution

The governing equations form a set of nonlinear differential and algebraic equations with θ and λ as variables. These equations were solved using an iterative approach. The governing equations for the free space state are (31) and (32). If we consider λ as a constant parameter then (31) becomes a second-order

Algorithm 1: Free Space Deformation.

Input: Membrane equations (ME), differential (DE) and algebraic (AE) beam equations, and boundary conditions.
Output: Slope θ .

```

1  repeat
2     $\theta_{s_j} \leftarrow 0$ ; {initial inclination}
3     $M_T \leftarrow$  Solve ME, nonlinear algebraic equations;
4     $\lambda_i \leftarrow 1$ ; {initial stretch}
5     $\theta_i, \theta'_i \leftarrow$  Solve DE (31);
6     $\lambda_{i+1} \leftarrow$  Solve AE (32);
7    repeat
8       $\theta_{i+1}, \theta'_{i+1} \leftarrow$  Solve DE (31);
9       $\lambda_{i+2} \leftarrow$  Solve AE (32);
10   until  $|\theta_{i+1} - \theta_i| \leq \text{error0}$ 
11    $\theta_{s_{j+1}} \leftarrow$  update  $\theta_s$  from the deformed geometry of
      beam
12 until  $|\theta_{s_{j+1}} - \theta_{s_j}| \leq \text{error2}$ 
13 return  $\theta_{i+1}$ 

```

Algorithm 2: Blocked Force State.

Input: ME, DE, AE, and boundary conditions.
Output: Slope θ and force Λ .

```

1   $\theta_{s_j} \leftarrow 0$ ; {here, inclination is assumed to be fixed}
2   $M_T \leftarrow$  Solve ME;
3   $\lambda_i \leftarrow 1$ ; {initial stretch}
4   $\theta_i, \theta'_i, \Lambda \leftarrow$  Solve DE (33), (43), and (47);
5   $\lambda_{i+1} \leftarrow$  Solve AE (44);
6  repeat
7     $\theta_{i+1}, \theta'_{i+1}, \Lambda \leftarrow$  Solve DE (33), (43), and (47);
8     $\lambda_{i+2} \leftarrow$  Solve AE (44);
9  until  $|\theta_{i+1} - \theta_i| \leq \text{error0}$ ;
10 return  $\theta_{i+1}, \Lambda$ ;

```

differential equation, with θ as a variable. Similarly, if θ is considered as a constant parameter, then (32) becomes a nonlinear algebraic equation, with λ as a variable. The solution procedure involves the following steps:

- 1) Solve (31) for $\lambda = 1$;
- 2) Solve (32) for θ and θ' obtained from the solution of (31);
- 3) Solve (31) for λ obtained from the solution of (32);
- 4) Repeat the above procedure until it meets a specified tolerance for θ .

The pseudocode of our solution algorithm is presented in Algorithm 1. A similar solution strategy was followed for the blocked force and grasping state. The Lagrange multiplier Λ is also an unknown parameter and is evaluated from the solution of governing differential equations. The pseudocodes for the block force and grasping state are given in Algorithms 2 and 3, respectively. In Algorithm 3, the pressure value for the initial guess is the minimum pressure required for the contact initiation between air chambers (see Supplementary Material, Section 1.3). If the solution does not converge within a defined number of iterations, then an upper limit of pressure $p_{\max}$ can be set to abort the solution.

Algorithm 3: Grasping State.

Input: ME, DE, AE, and boundary conditions.
Output: Slope θ and pressure p .

```

1   $p_j \leftarrow p$ ; {initial guess pressure}
2  repeat
3     $\theta_{s_j} \leftarrow 0$ ; {here, inclination is assumed to be fixed}
4     $M_T \leftarrow$  Solve ME;
5     $\lambda_i \leftarrow 1$ ; {initial stretch}
6     $\theta_i, \theta'_i, \Lambda \leftarrow$  Solve DE (33), (43), and (47);
7     $\lambda_{i+1} \leftarrow$  Solve AE (44);
8    repeat
9       $\theta_{i+1}, \theta'_{i+1}, \Lambda \leftarrow$  Solve DE (33), (43), and (47);
10      $\lambda_{i+2} \leftarrow$  Solve AE (44);
11   until  $|\theta_{i+1} - \theta_i| \leq \text{error0}$ ;
12    $p_j \leftarrow p_j + \Delta p$ ;
13   until  $|F_{ext} - \Lambda| \leq \text{error1}$  or  $p_j \geq p_{\max}$ ;
14   return  $\theta_{i+1}, p_j$ ;

```

IV. SIMULATIONS AND EXPERIMENTS

A. Material Properties

The theoretical model is based on incompressible neo-Hookean material, and it requires the shear modulus as an input parameter. Tension tests were performed for Dragon Skin (DS) 20 and DS 30 silicone rubber using a tensile tester (EMX-275, IMADA, Japan). The test setup is shown in the inset of Fig. 5(a), and the dog bone sample used in the tests is shown in the inset of Fig. 5(b). Four samples with a thickness of 2.25 mm were fabricated using casting process. The tests were conducted at a speed of 100 mm/min. The Young's modulus E is obtained from a linear fit to the stress strain data, and the shear modulus μ is calculated using the relation

$$\mu = \frac{E}{2(1 + \nu)}. \quad (48)$$

B. Frictional Properties

The frictional properties of the actuator material are required for both the analytical and FE simulations. The COF was obtained by conducting experiments on a friction measuring machine (TL201 Tt, Trinity-Lab. Inc., Japan). The test setup is shown in Fig. 6(a). A V-shaped test specimen fabricated using DS material was used to replicate the tip contact conditions. The COF was measured for the contact between DS versus DS silicone rubber, and for DS versus Polylactic Acid (PLA) materials. The friction test data is presented in Fig. 6(b). In simulations, we considered the COF value corresponding to 0–0.5 N, because the force magnitude in the grasping simulations was in this range.

C. Analytic Solutions

The derived governing equations were solved numerically using “MATLAB R2019b.” The differential equations were solved using “bvp5c, boundary value problem solver” while the algebraic equations were solved using “fsolve, nonlinear equation solver.” The integral equations (45) and (46) were solved

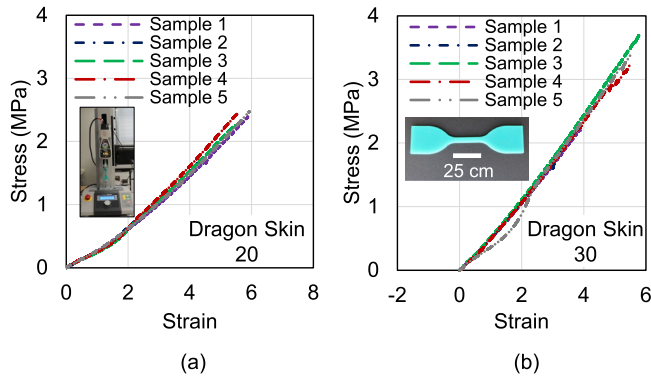


Fig. 5. Stress and strain curve for (a) DS 20 and (b) DS 30 silicone rubber.

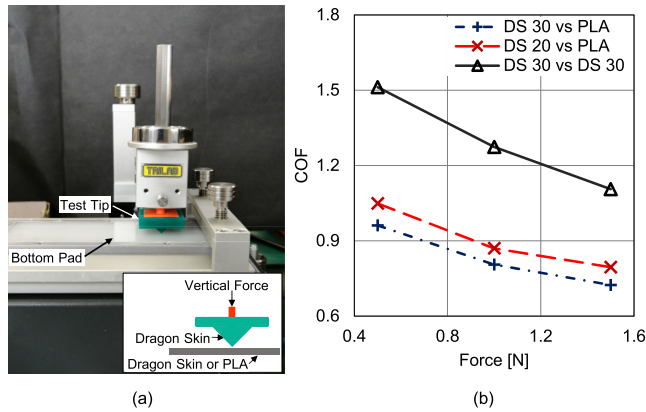


Fig. 6. Coefficient of friction measurement. (a) Experimental setup and schematic of friction tester tip and (b) friction test data for DS 30 versus PLA, DS 20 versus PLA, and DS 30 versus DS 30.

using “cumtrapz, cumulative trapezoidal numerical integration.” The equations were solved for 120 discrete node points. The error tolerances “error0,” “error1,” and “error2” were set to 0.001, 0.00001, and 0.00001, respectively. Computations were run on a desktop computer [Dell Optiplex; Windows 10 Pro; Intel Core i7-9700 CPU @3.00 GHz $\times$ 64; 8 GB RAM]. For the prescribed error tolerances, the solver usually requires three to ten iterations to determine a solution. The accuracy of the solution improves with an increase in the number of nodes, but after 30 nodes, the variation in the solution was very small (less than 1% of the solution value). The computation time for the solution was usually less than 60 s. Approximately 60%–90% of the computation time was consumed by the “bvp5c” solver. To calculate the deformation of the actuator, the solvers have to solve the two governing equations. This is the main reason for the lower computation time requirement of the model when compared with the FE-based model, which requires several hours to predict the deformation of the actuator subjected to similar loading conditions. The governing algebraic equation calculates the axial stretch induced in the actuator. The maximum recorded axial stretch for the bending behavior of the actuator was less than 10%. The variation in the tip coordinates of the actuator was less than 5% when calculated without considering the axial stretch. In finger-based grippers, bending deformation is the primary objective; hence, the axial stretching effect is optional.

TABLE II
ACTUATOR MATERIAL AND GEOMETRIC PARAMETERS

Actuator Type	A	B	C	D
Shore A Hardness	20	30	20	30
Color	Green	Aquamarine	White	Yellow
E (kPa)	409	595	409	595
μ (kPa)	136	198	136	198
$2h_x$ (mm)	1.8		1.5	
t_m (mm)	1.1		1.5	
h_L (mm)	4.2		4.5	
$2a$ (mm)	10		10	
$2b$ (mm)	16		16	
H (mm)	5		5	
B (mm)	20		20	
L (mm)	77		77	

The computation time depends on the number of iterations. For large deformation gradients, the “bvp5c” solver required more iterations to converge. Large deformation gradients occur when the actuator is subjected to forces and moments of higher magnitude. At a higher number of nodes, the solution does not satisfy the solver’s internal function tolerances, which results in premature termination of the solver. The numerical solution of a differential equation depends on the initial guess for the solution. For the first iteration, the initial guess was set to $\theta = 0$ and $\theta' = 0$, and from the second iteration, the solution of the previous iteration was considered as the initial guess.

Analytical computations were performed for three different states of the actuator: 1) free space; 2) blocked force; and 3) grasping state.

1) *Free Space Deformation*: The undeformed state of the actuator was considered horizontal during the simulation. The simulations were performed for four actuators made of different hardness materials and geometries. The geometric and material parameters of the actuators are listed in Table II. The Poisson’s ratio ν was taken as 0.50. Here, gravity was considered as distributed forces: $g_x = 0$ N/mm and $g_y = -0.002$ N/mm.

2) *Blocked Force*: Simulations for the blocked force state were performed to predict the tip force generated by the actuator. The governing equations for the tip contact state were considered, and the boundary conditions were modified as follows: $P = 0$ and $v_0 = y_c = 0$, at $X = L$. The objective of this simulation is to calculate the Lagrange multiplier Λ , which is the force generated at the tip along the Y -direction.

3) *Grasping State*: Simulations for the grasping state were performed to predict the air pressure required for a stable grasp. These simulations were performed on a three-finger gripper to grasp a hexagonal cylinder (side 15 mm and height 80 mm), circular cylinder (radius 18 mm and height 80 mm), 3-face cone (base side 50 mm and height 80 mm), and right circular cone (radius 30 mm and height 80 mm). The pinching mode for grasping was considered in the simulations. The undeformed state of the actuator was considered to be vertical during the simulation. Here, gravity was considered as distributed forces: $g_x = 0.002$ N/mm and $g_y = 0$ N/mm. The boundary condition at $X = L$ was modified as follows: $P = H_{\text{ext}} = m_h g/3$. The force value vertical to the actuator (tip force) required for a

stable grasp was obtained using Coulomb's law of friction as $F_{\text{ext}} \geq H_{\text{ext}}/\mu_f$. For 3-face and circular cones, it was obtained as: $F_{\text{ext}} \geq H_{\text{ext}}/(\mu_f \sin^2 \beta)$, where β is the base angle of the 3-face and circular cones.

D. FE Simulations

FE models for simulating the free space and block force states were constructed using Abaqus, 2017 (Dassault Systèmes, MA). The CAD designs of the actuators were input into Abaqus and meshed with 10-node quadratic tetrahedron elements (C3D10) and the global mesh seed was set to 1mm. Gravity was included in the simulations and the material densities were set to 1080 kg/m³ for DS 20 and DS 30 according to the material datasheet. The same Young's modulus given in Table II was used in the material property settings. In the block force state, a force sensor was modeled as a cubic part and meshed with 8-node linear hexahedron elements (C3D8R). The contact interactions between the air chambers and between the actuator and the force sensor were modeled using the penalty method with the measured COF. The geometrical nonlinearity was included in the model to account for large deformation. Standard/static simulations were conducted, and the simulated deformation is shown in the Supplementary Material, Section-2. The reaction force in the vertical direction on the sensor was recorded as the block force and compared with the experimental measurements.

E. Experiments

The actual deformation and force characteristics of the actuators were captured for free space, blocked force, and grasping states. Air pressure was generated using an off-board compressor (JUN-AIR 3-4) and controlled using a digital pressure regulator (SMC ITV2030). The deformation of the actuators was captured by photo snapshots using a camera (Canon Inc.). The camera images were processed for lens distortion using "MATLAB R2019b Single Camera Calibrator App." The nodal coordinates of the actuator in the deformed state were calculated using ImageJ software.¹ In the blocked force state, the actuator was initially horizontal, and the free end of the actuator was resting on a 3-axis force sensor (USL06-H5 Force Sensor, Tech Gihan), which was rigidly attached to the base, as shown in Fig. 7(c). Grasping tests were performed using three-finger grippers to grasp a hexagonal cylinder, circular cylinder, 3-face cone, and right circular cone. These experiments were performed to validate the range of air pressures predicted by the proposed model for stable grasps.

V. RESULTS AND DISCUSSION

A. Free Space Deformation

The experimental snapshots of free space deformation at different air pressures are shown in Fig. 8. The experimental, analytical, and FE simulated free space deformation plots of the actuators at different air pressures are shown in Fig. 9. The deformation predicted by the analytical model is close to

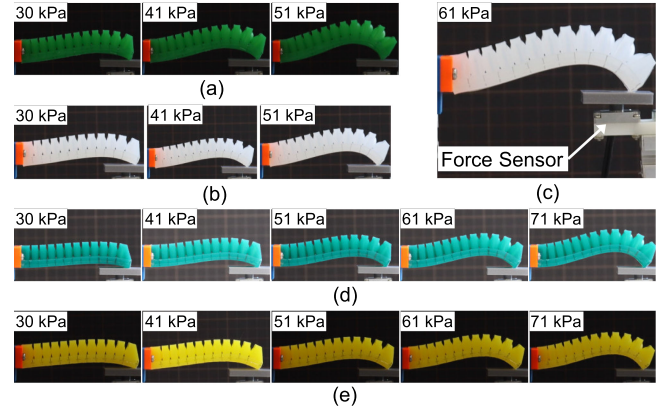


Fig. 7. Experimental snapshots of block force deformation of (a) type A, (b) type C, (c) type C, (d) type B, and (e) type D actuators.

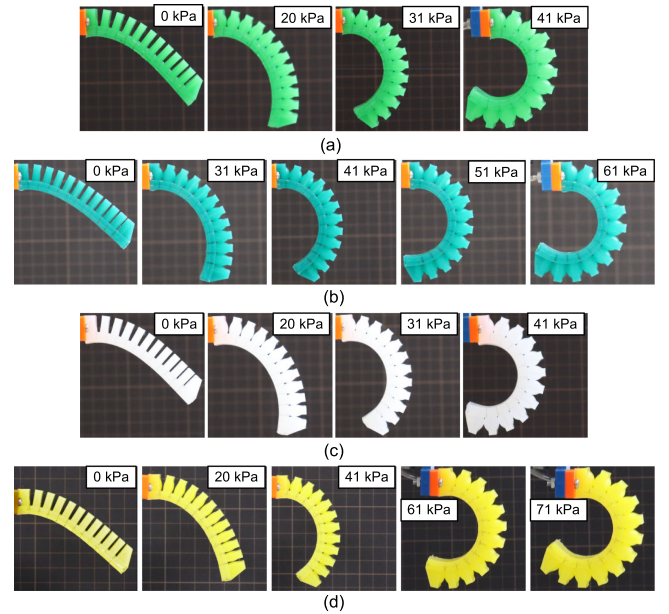


Fig. 8. Experimental snapshots of free space deformation of (a) type A, (b) type B, (c) type C, and (d) type D actuators.

the experimental deformation of the actuators, while the FE simulations overestimate the deformation. The upper limit of air pressure for the analytical model is the same as that of the FE model for type A [see Fig. 9(a)] and C [see Fig. 9(c)] actuators but it is higher for type B [see Fig. 9(b)] and D [see Fig. 9(d)] actuators. The FE model did not predict the deformation of type B and D actuators beyond 51 and 61 kPa, respectively. While the theoretical model predicts up to 61 and 71 kPa for type B and D actuators, respectively. The accuracy of the analytical model for estimating the free space deformation is better than that of the FE model. Near the clamped end of the actuators, the predicted deformation from the theoretical model is greater than the experimental results and FE simulations. This occurs because of the additional stiffness provided by the air chambers to the actuator cover. The analytical and FE simulations were also carried out for actuators having different length (Supplementary Material, Section-6). The model is further implemented to predict the free

¹[Online]. Available: <https://imagej.nih.gov/ij/>

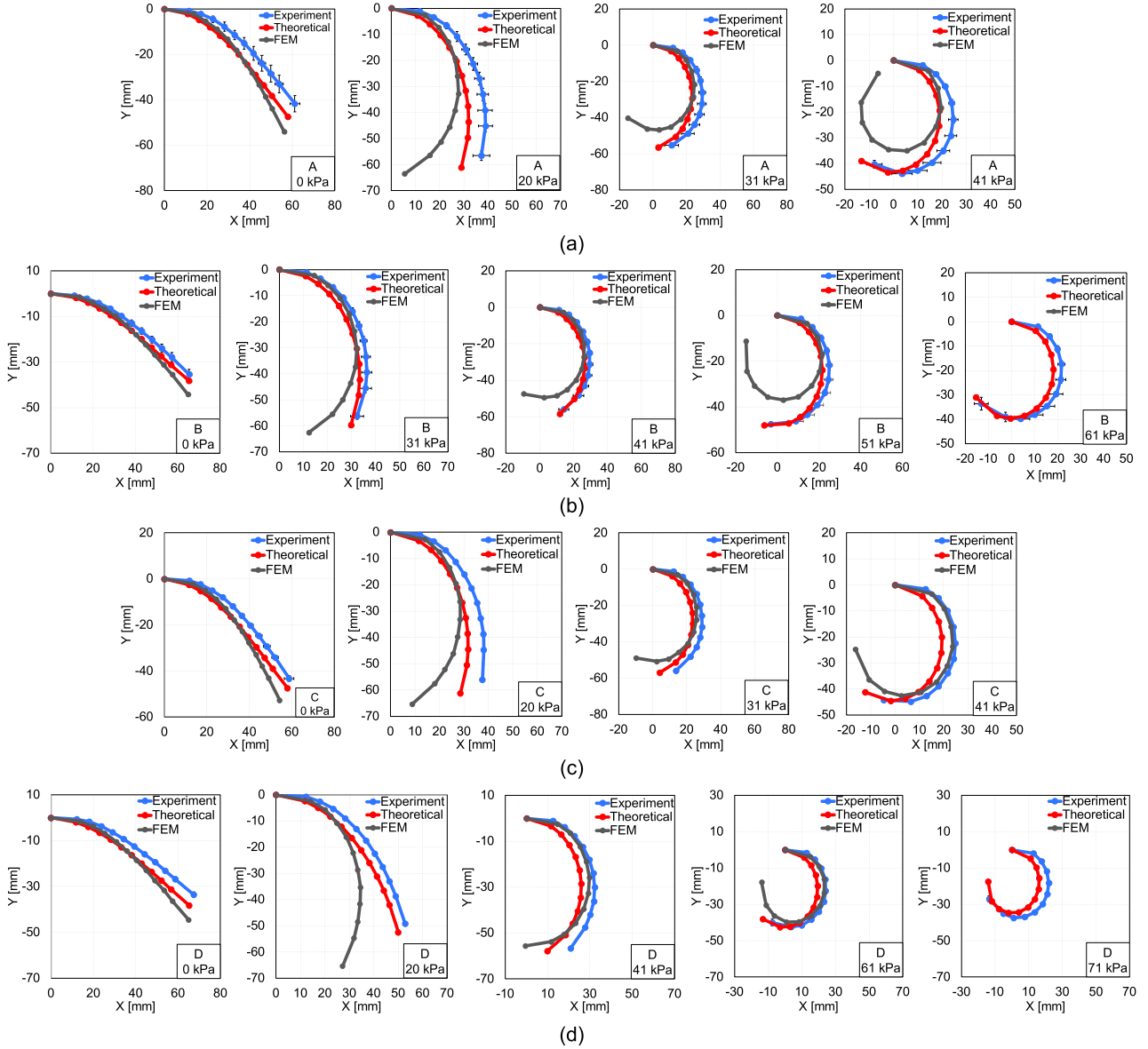


Fig. 9. Theoretical, FEM, and experimental free space deformation plots of (a) type A, (b) type B, (c) type C, and (d) type D actuators. Error bars indicate the standard deviation among nine trials.

space deformation of a nonuniform chamber geometry actuator. Refer to the Supplementary Material for free space deformation of a nonuniform geometry actuator, Section-5.

B. Blocked Force

The experimental snapshots of the blocked force deformation of the actuators are shown in Fig. 7. The experimental, analytical, and FE simulated blocked force plots of the actuators are shown in Fig. 10. The force results are presented for 30 kPa to 70 kPa air pressure. At air pressure below 30 kPa, the force generated by the actuators is very low in magnitude (i.e., less than 0.1 N) and it was hard to measure using our force sensor. The analytical and FE models can predict the block force up to 41 and 30 kPa, respectively, for type A [see Fig. 10(a)], and up to 51 kPa for type C actuator [see Fig. 10(c)]. The analytical and FE models

predict the blocked force up to 71 kPa for type B [see Fig. 10(b)] and 61 and 71 kPa, respectively, for type D [see Fig. 10(d)] actuator. The material of type A and C actuators was less stiff than type B and D actuators. Owing to the large deformation gradient value of the air chambers, the solvers stopped prematurely at pressure magnitudes beyond this limit. For the block force deformation plots, refer to the Supplementary Material, Section-4. The analytical computations and experiments were further performed for different contact heights at the tip. Refer to the Supplementary Material, Section-3, for the force results for different contact height.

C. Grasping State

The theoretical predicted range of air pressure required for a stable grasp of different objects is listed in Table III. The

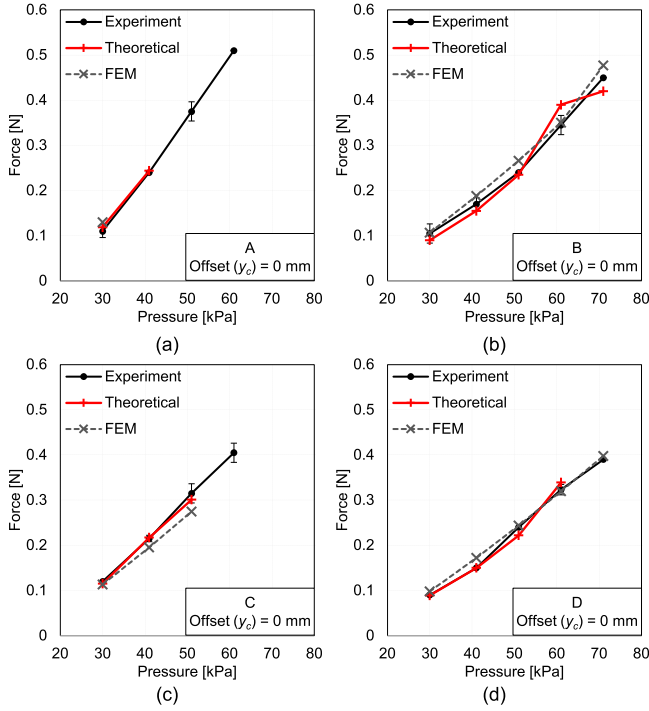


Fig. 10. Theoretical, FEM and experimental block force plots of (a) type A, (b) type B, (c) type C, and (d) type D actuators. Error bars indicate the standard deviation among two trials.

TABLE III
THEORETICAL PREDICTED RANGE OF AIR PRESSURE

Grasping Objects	Mass m_h (g)	Pressure required by actuators p (kPa)			
		A	B	C	D
Hexagonal Cylinder	35	—	42	37	43
Circular Cylinder	26	28	32	28	32
3-Face Cone	19	28	33	29	33
Circular Cone	20	25	29	25	29

COF between the grasping objects (PLA) and the actuator was taken as 0.96. Snapshots of the grasping experiment for type A actuator are shown in Fig. 11. The gripper was able to lift the circular cylinder and 3-face cone at the theoretically predicted pressure. The type B gripper (see Fig. 12) was able to lift the hexagonal and circular cylinders. The type C gripper (see Fig. 13) was able to lift the hexagonal cylinder, circular cylinder, and 3-face cone. The type D gripper (see Fig. 14) was able to lift the hexagonal and circular cylinders. However, the grasps were unstable. The theoretically predicted pressure is the minimum pressure required for grasping. In the experiments, we did not consider any factor of safety for the predicted air pressure. If a certain magnitude factor of safety is considered, then a stable grasp can be attained by the actuators. In addition, the predicted air pressure is for the pure static case, and in the experiments, we moved the gripper vertically to lift the objects. This dynamic effect is also a factor contributing to the unstable grasp.

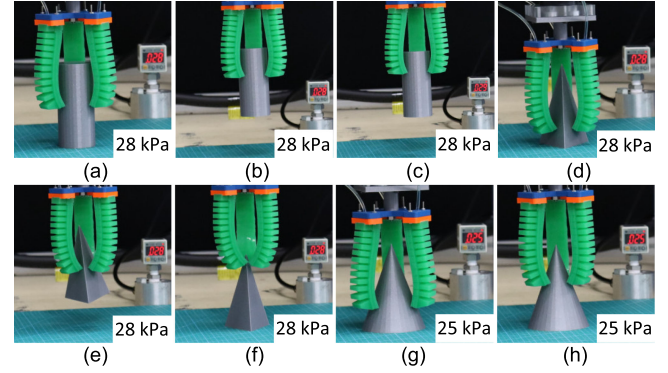


Fig. 11. Photo snapshots of grasping experiment for type A actuator. The experiments were performed to grasp a circular cylinder (a)–(c), 3-face cone (d)–(f), and circular cone (g) and (h).

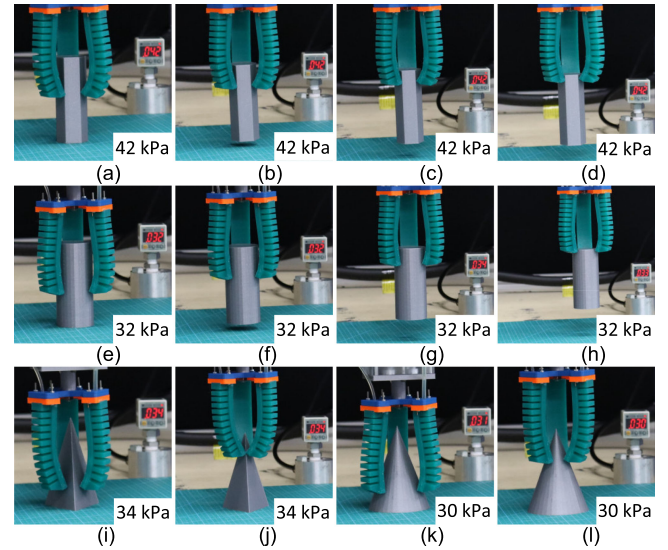


Fig. 12. Photo snapshots of grasping experiment for type B actuator. The experiments were performed to grasp a hexagonal cylinder (a)–(d), circular cylinder (e)–(h), 3-face cone (i) and (j), and circular cone (k) and (l).

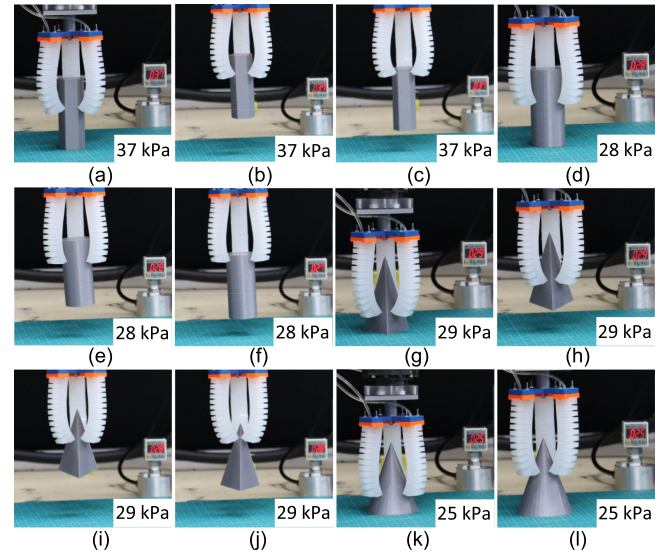


Fig. 13. Photo snapshots of grasping experiment for type C actuator. The experiments were performed to grasp a hexagonal cylinder (a)–(c), circular cylinder (d)–(f), 3-face cone (g)–(j), and circular cone (k) and (l).

TABLE IV
COMPARISON WITH EXISTING MODELS

Model	Cover Modeling	Chamber Modeling	Material Model	Output
Alici <i>et al.</i> [30]	Euler–Bernoulli beam based constant curvature	—	Linear	Deformation, Force
Zhong <i>et al.</i> [33]	Euler–Bernoulli beam based constant curvature	—	Linear	Deformation
Cao <i>et al.</i> [44]	Euler–Bernoulli beam based constant curvature	—	Linear	Deformation
Liu <i>et al.</i> [45]	Serially connected constant curvature	Membrane based	neo-Hookean	Deformation, Force
Wang <i>et al.</i> [47]	Serial robot	Surface based	Yeoh	Deformation, Force, Dynamics
Fang <i>et al.</i> [48]	Fitting geometric profile	Fitting geometric profile	Linear	Deformation
Xu <i>et al.</i> [49]	Finite element based approach	Finite element based approach	Yeoh	Deformation
Payrebrune <i>et al.</i> [50]	Rod theory	—	Linear	Force, Deformation
Gu <i>et al.</i> [51]	Rod theory	—	Linear	Force, 3D-deformation, multi-material
Wang <i>et al.</i> [52]	Optimization based parameter identification method	—	Linear	Deformation, Force, Dynamics
Current model	Finite strain Euler–Bernoulli beam	Finite strain membrane theory	neo-Hookean	Deformation, Force

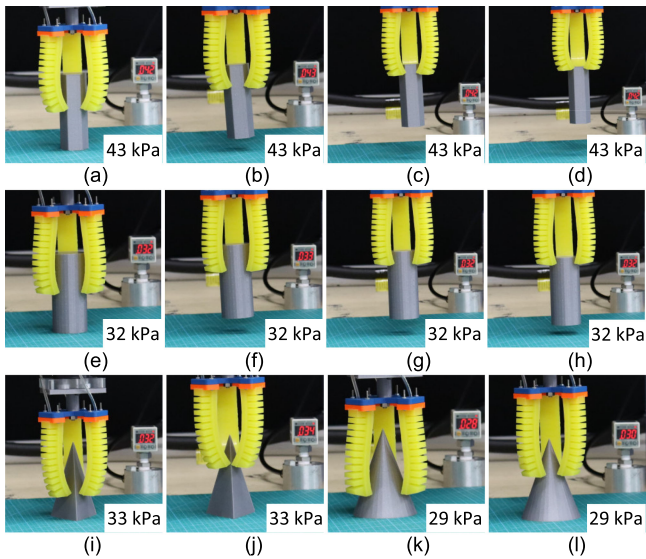


Fig. 14. Photo snapshots of grasping experiment for type D actuator. The experiments were performed to grasp a hexagonal cylinder (a)–(d), circular cylinder (e)–(h), 3-face cone (i) and (j), and circular cone (k) and (l).

The 3-face and right circular cones have a tapered geometry, which was considered while estimating the required pressure range, but the grippers failed to lift the circular cone. This occurs because of the contact condition between the surface of the cone and the actuator tip. A full edge contact of the actuators occurred at the surface of the 3-face cone, whereas full edge contact did not occur for the circular cone. Moreover, the COF for an inclined surface might be different from that of a planar surface.

VI. CONCLUSION

The proposed analytical model predicted the deformation and force characteristics of a pneu-net soft actuator. The model considered the axial stretch, which was generally not considered in bending actuators in other studies; however, it was an important aspect for gait-type soft robots. The modeling of air chambers was based on the finite strain membrane theory and the contact mechanics approach. It could be adapted to model other soft actuators such as flat membrane-based shell gripper. In the modeling of air chambers, the proposed model considered the

geometric variations in the design of air chambers. Apart from the geometry of air chambers, the actuators used for grasping applications generally had a flat base. Hence, the proposed rectangular cross-sectional beam-based model could be applied to the majority of such actuators. Theoretically, a similar modeling approach could be implemented for other actuators with a different geometric design other than the pneu-net actuator, such as cylindrical and ribbed-type actuators [1]. By modifying the air chamber model, the actuators could be modeled using a similar approach. When compared with the constant curvature models that predicted either deformation or force characteristics, the proposed model predicted the deformation and force characteristics of the actuator simultaneously for the grasping state. A brief comparison of the proposed model with the existing models of pneu-net actuators was presented in Table IV. The theoretical formulation of proposed model was different from the existing models. The proposed model was efficient in terms of computation time because it consisted of nonlinear differential and nonlinear algebraic equations as the final governing equations. The proposed solution strategy was effective for solving the equations numerically, using a boundary value problem and a nonlinear algebraic equation solver. The presented work also provided an insight on the frictional properties at the actuator tip and grasping object contact interface.

The presented model was more abstract and did not require a CAD file for simulation, it only required the geometric and material parameter values. It could be used with other optimization techniques to find the best-fit design parameters from a large design space (i.e., initial design) as it was more than 100 times faster than the FE model and did not need a new CAD file for each iteration. In the presented model, it was easy to tune the material parameters. As an example, we could tune the material parameters for free space deformation and used them to predict the force behavior. In the simulations, we used the experimentally calculated material parameters. But if we tune the deformation behavior by varying the material parameters then we can further reduce the model prediction error. In applications, the proposed model could be used to predict the deformation and force characteristics, and to predict the air pressure required for a stable grasp by a gripper. Here, we considered only distributed and terminal forces, but a similar modeling approach could be followed for nonterminal forces. In this study, we considered

a three-finger gripper. The same model can be used for two, four, or more than four fingers by modifying the boundary conditions and transformation matrices for global coordinates. We considered standard geometry grasping objects, but the model can be applied to grasp arbitrarily shaped objects if the outer boundary and weight of the objects are known. In the presence of large deformation gradient, the solver terminated prematurely. This occurred because of the solver's internal function tolerances, which resulted in an unexpected solution. The theoretical model failed to predict the air pressure required for a stable grasp of the circular cone even after considering the tapered geometry at the contact. This was due to the lesser edge contact of the actuator tip at the circular cone surface when compared with the contact at the 3-face cone surface.

From soft robotics perspective, the presented work enriched the modeling of soft robots and provided another way to model soft pneu-net actuator. With the help of a modeling tool, we can explore larger design space. The presented work highlighted the application of large strain beam and large strain membrane theories to soft robots. The presented design approach could be extended to facilitate the design of other soft robots. The future work could involve an investigation of the behavior of the actuator for various contact conditions while grasping soft objects. The model could also be further extended to study the dynamic behavior of the actuator. The membrane model for the air chambers would be further applied to study the behavior of flat membrane-based shell gripper.

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